Alex Rivera

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EDUCATION

State University Expected Graduation: **May 2027**
Bachelor's Degree in Computer Science; Minor in Statistics
Relevant Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning

EXPERIENCE

Software Engineering Intern *June 2026 – August 2026*
Acme Corp *San Jose, CA*

- Built a Python service that batches inventory updates into PostgreSQL from 40+ warehouses, cutting nightly sync time from 3 hours to 25 minutes.
- Designed and shipped 6 REST endpoints in TypeScript and Node.js for the order-tracking dashboard, with request validation and 92% unit test coverage.
- Wrote SQL data-quality checks that flag duplicate and late inventory events, surfacing about 1,200 bad rows a week for the operations team.
- Added a Docker-based integration test stage to the team's CI pipeline that caught 3 schema regressions before release during my internship.
- Documented the sync service's failure modes in a runbook and presented it to 12 engineers at the team's weekly review.

Undergraduate Research Assistant *January 2026 – Present*
Northwind Labs, State University *Oakland, CA*

- Train convolutional image classifiers in PyTorch to detect crop disease from field photos, reaching 88% validation accuracy on 9 disease classes.
- Built a labeling pipeline that de-duplicates and splits 15,000 field images by farm, preventing train/test leakage between nearby photos.
- Evaluate models with per-class precision and recall and a held-out farm split, and report results to the lab in a biweekly experiment log.
- Co-wrote a 4-page workshop paper draft on low-cost crop disease detection with a PhD mentor.

PROJECTS

Replicated Key-Value Store (Go) | *Course Project* *Fall 2025*

- Implemented Raft leader election and log replication in Go for a 5-node key-value store that stays available with 2 nodes down.
- Tested the store under injected network partitions and crashes with a randomized harness that ran 1,000 trials per change.
- Batched log entries and pipelined append requests, raising write throughput from 1,800 to 6,500 operations per second on a 5-node local cluster.

Campus Chat (React, Node.js, Redis) | *github.com/alexrivera/campus-chat* *Spring 2026*

- Used Redis pub/sub to fan out messages across 3 Node.js server instances, keeping delivery under 100 ms in load tests with 500 simulated users.
- Built a real-time chat app in React and TypeScript with WebSocket rooms, typing indicators, and message history for a 300-member student club.
- Wrote 140 Jest and Playwright tests and ran them in GitHub Actions on every pull request.
- Added email sign-in with one-time codes and per-room permissions, so moderators can mute users and lock rooms without a server restart.

Leaf Doctor (Python, PyTorch, FastAPI) | *github.com/alexrivera/leaf-doctor* | *live demo* *Summer 2025*

- Served the model behind a FastAPI endpoint in Docker that returns a diagnosis and confidence score in under 300 ms.
- Fine-tuned a ResNet-18 on 20,000 public plant images to classify 12 leaf diseases with 91% test accuracy.
- Compared 4 augmentation strategies in a tracked experiment grid and kept the one that raised accuracy on blurry phone photos from 74% to 85%.

SKILLS

Languages: Go, Python, SQL, Java, TypeScript
Frameworks & Tools: PostgreSQL, Docker, GitHub Actions, Redis, Node.js, pytest, Git
Concepts: REST APIs, Distributed Systems, Consensus (Raft), CI/CD, Unit Testing